1.2 Ordinary Differential Equations

2 Stability and accuracy of the numerical methods

2.1 Stability

Question 1.

We run our program with the given initial parameters.

n	x_n	Y_n	$y(x_n)$	E_n	G_n	ΔG_n
0	0	0	0	0		
1	0.6	1.8	0.45809	1.3419	0.49015	
2	1.2	-1.5321	0.29296	-1.8251	1.0027	0.51258
3	1.8	2.6871	0.16455	2.5226	1.5421	0.53942
4	2.4	-3.4645	0.09065	-3.5551	2.114	0.57183
5	3	5.0135	0.049781	4.9638	2.6703	0.55629
6	3.6	-6.9293	0.027323	-6.9567	3.2328	0.56256
7	4.2	9.7503	0.014996	9.7353	3.7929	0.56009
8	4.8	-13.6234	0.0082297	-13.6316	4.354	0.56106
9	5.4	19.0875	0.0045166	19.083	4.9147	0.56068
10	6	-26.7144	0.0024788	-26.7169	5.4755	0.56083
11	6.6	37.4047	0.0013604	37.4033	6.0363	0.56077
12	7.2	-52.3641	0.00074659	-52.3648	6.5971	0.56079
13	7.8	73.311	0.00040973	73.3106	7.1578	0.56078
14	8.4	-102.6347	0.00022487	-102.6349	7.7186	0.56079
15	9	143.689	0.00012341	143.6889	8.2794	0.56079

Table 1: Euler method with $h = 0.6$

We observe that E_n oscillates between positive and negative values, as expected. Given that $|E_n|$ grows proportional to $e^{\gamma x}$, we see that $\frac{1}{h}\log|E_n| \approx \gamma n + C$ for some constant C. Calculating $G_n \equiv \frac{1}{h}\log|E_n|$ and $\Delta G_n \equiv G_n - G_{n-1}$ for each value of n, we see that $\gamma \approx 0.561$.

We run our program again with different values of h, noting down the last three values of E_n and calculating ΔG_N as an estimate for the growth rate γ. Here $N \equiv 9/h$ (so that $y(x_N) = y(9)$, as above).

h	Y_N	E_N	E_{N-1}	E_{N-2}	ΔG_N
0.1	0.00012145	$-1.9584 \times 10{-}6$	-2.1643×10^{-6}	-2.392×10^{-6}	-1
0.3	0.00011806	-5.3542×10^{-6}	-7.2275×10^{-6}	-9.7561×10^{-6}	-1
0.45	-0.010711	-0.010834	0.013521	-0.016935	-0.49239
0.5	-0.93357	-0.9337	0.93368	-0.93371	4.643×10^{-5}
0.6	143.689	143.6889	-102.6349	73.3106	0.56079
0.9	-12677.2767	-12677.2768	4875.8757	-1875.3369	1.0617
1.5	-13461.7552	-13461.7553	2692.351	-538.4706	1.073
1.8	7772.0111	7772.011	-1253.5503	202.1848	1.0136

Table 2: Effect of step size h on the size of the instability

We observe that, for $h \leq 0.9$, decreasing the step size h causes the size of the instability and the growth rate γ to decrease, with $\gamma \approx 0$ for $h = 0.5$ and $\gamma < 0$ for $h \leq 0.45$ (indicating that the absolute value of the error decreases with each step). γ appears to approach a minimum value of -1 as h approaches 0. Also, for sufficiently small values of h, the error no longer oscillates between positive and negative values. Interestingly, γ does not increase monotonically with h, but appears to reach a maximum somewhere in the region $0.9 \leq h \leq 1.8$.

The explanation for this behaviour as h varies will be explored further in Question 2.

Question 2.

(i) We can rewrite equation (7) as follows.

$$Y_{n+1} - (1-4h)Y_n = 3he^{-nh} \tag{I}$$

Any solution Y_n to equation (I) can be written as the sum of a complementary function $Y_n^{(c)}$ and a particular integral $Y_n^{(p)}$.

The complementary function to the homogeneous equation $Y_{n+1} - (1-4h)Y_n = 0$ is given by

$$Y_n^{(c)} = A(1-4h)^n$$

For the particular integral, we try a solution of the form $Y_n^{(p)} = Be^{-nh}$. Substituting this into equation (I) gives

$$B(e^{-h} - 1 + 4h)e^{-nh} = 3he^{-nh}$$

$$\implies B = \frac{3h}{e^{-h} - 1 + 4h}$$

Imposing the initial condition

$$Y_0 = Y_0^{(p)} + Y_0^{(c)} = 0$$

tells us that $A = -B$, and so the analytic solution to the difference equation is given by

$$Y_n = \frac{3h}{e^{-h} - 1 + 4h}[e^{-nh} - (1-4h)^n] \tag{II}$$

(ii) We have $E_n = Y_n - y(x_n) = Y_n^{(c)} + Y_n^{(p)} - y(x_n)$. For the rest of Question 2(ii), we write "$n \to \infty$" as shorthand for "the limit as $n \to \infty$, holding h constant". We observe that the analytic solution (6) to the original differential equation has $y(x) \to 0$ as $x \to \infty$. Similarly, the particular integral $Y_n^{(p)} = \frac{3h}{e^{-h}-1+4h}e^{-nh} \to 0$ as $n \to \infty$. However, if $|1-4h| \geq 1$, the complementary function

$$Y_n^{(c)} = \frac{3h}{e^{-h} - 1 + 4h}(1-4h)^n$$

does not tend to zero, and in fact diverges if $|1-4h| > 1$, resulting in the observed instability in the numerical solution Y_n. Since $h > 0$, this instability occurs when $h > \frac{1}{2}$, which gives $(1-4h) < -1$.

For $h > 1/2$ and large n, since both $Y_n^{(p)}$ and $y(x_n)$ tend to zero, $E_n = Y_n^{(p)} + Y_n^{(c)} - y(x_n) \approx Y_n^{(c)}$. Then, since $(1-4h)$ is negative, the sign of $Y_n^{(c)}$ alternates between $+$ for even n and $-$ for odd n, resulting in the observed oscillation in $\mathrm{sgn}(E_n)$. Also, $E_n \approx Y_n^{(c)}$, which is proportional to $(1-4h)^n = (1-4h)^{x_n/h}$. Then we see that the growth rate γ is given by

$$\gamma = \frac{1}{h}\log|1-4h| \tag{III}$$

We can compare the value of γ given by equation (III) with the values of G_N from Table 2.

h	γ	ΔG_N
0.45	-0.495875	-0.49239
0.5	0	4.643×10^{-5}
0.6	0.560787	0.56079
0.9	1.061679	1.0617
1.5	1.072959	1.073
1.8	1.013638	1.0136

Table 3: Theoretical value of γ and calculated value of ΔG_N for selected values of h

In the case $h = \frac{1}{2}$, $(1-4h) = 1$, so $Y_n^{(c)} =$ const. and we expect the error E_n to approach a constant value, hence $\gamma = 0$.

For $h < 1/2$, $|1-4h| < 1$, so we expect the error to decay as $n \to \infty$, and we expect γ to be negative. From equation (6), $y(x_n) = O(e^{-nh})$ as $n \to \infty$. Also, from the derivation in (i), we see that $Y_n^{(p)} = O(e^{-nh})$ and $Y_n^{(c)} = O((1-4h)^n)$ as $n \to \infty$.

Then, if $|1-4h| > e^{-h}$, the $Y_n^{(c)}$ term dominates, and for large n, $E_n \approx Y_n^{(c)}$, so equation (III) above still holds. Numerically solving $|1-4h| = e^{-h}$, we see that this is true for $h > 0.415...$.

Else, $|1-4h| \leq e^{-h}$. If $|1-4h| < e^{-h}$, then for large n,

$$E_n \approx \left[\frac{3h}{e^{-h} - 1 + 4h} - 1\right] e^{-nh}$$

and we see that $\gamma = -1$, as in Table 2.

In the boundary case $|1-4h| = e^{-h}$, for large n,

$$E_n \approx \frac{3h}{e^{-h} - 1 + 4h}[(e^{-h})^n - (-e^{-h})^n] - e^{-nh}$$

In this case, E_n does not decay monotonically, but we still have $E_n = O(e^{-nh})$.

(iii) Using $x_n \equiv nh$, we rewrite our expression (II) for Y_n in terms of x_n and h only.

$$Y_n = \frac{3h}{e^{-h} - 1 + 4h}[e^{-x_n} - (1-4h)^{-4x_n/-4h}]$$

We take the limit $h \to 0$ while keeping x_n fixed. Observe that

$$\begin{aligned}\lim_{h \to 0} \frac{3h}{e^{-h} - 1 + 4h} &= \lim_{h \to 0} \frac{3h}{1 - h + O(h^2) - 1 + 4h} \\ &= 1\end{aligned}$$

Also,

$$\begin{aligned}\lim_{h \to 0}(1-4h)^{1/-4h} &= \lim_{k \to 0}(1+k)^{-4/k} \\ &= e\end{aligned}$$

Rewriting $x_n = x$ and substituting these into our original expression yields

$$Y = e^{-x} - e^{-4x}$$

and we recover the analytic solution (6) to the original differential equation.

2.2 Accuracy

Question 3.

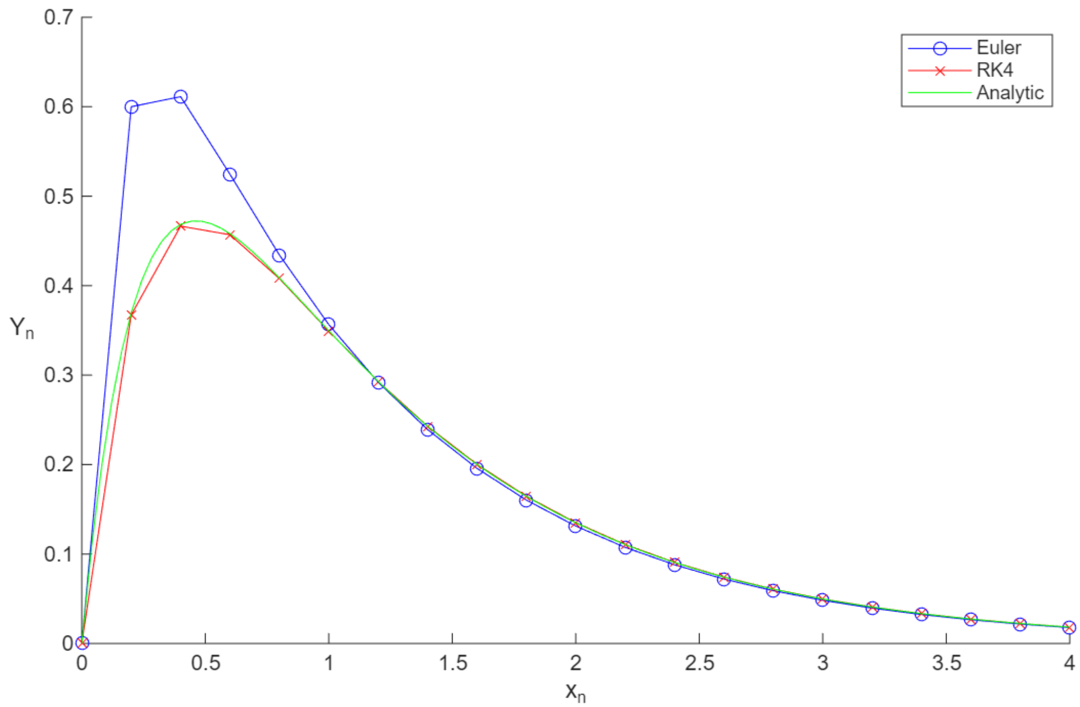

Figure 1: Y_n against x_n using the Euler (blue) and RK4 (red) methods with analytic solution (green)

Question 4.

k	h	E_n (Euler)	E_n (RK4)
0	0.4	0.73158	-0.062568
1	0.2	0.14281	-0.0019233
2	0.1	0.063716	-8.4201×10^{-5}
3	0.05	0.030039	-4.4134×10^{-6}
4	0.025	0.014599	-2.5268×10^{-7}
5	0.0125	0.0071983	-1.5117×10^{-8}
6	0.00625	0.0035744	-9.2435×10^{-10}
7	0.003125	0.0017811	-5.7144×10^{-11}
8	0.0015625	0.00088901	-3.5521×10^{-12}
9	0.00078125	0.00044413	-2.2140×10^{-13}
10	0.00039063	0.00022197	-1.3819×10^{-14}
11	0.00019531	0.00011096	-8.6307×10^{-16}
12	9.7656×10^{-5}	5.5474×10^{-5}	-5.3923×10^{-17}
13	4.8828×10^{-5}	2.7736×10^{-5}	-3.3696×10^{-18}
14	2.4414×10^{-5}	1.3868×10^{-5}	-2.1058×10^{-19}
15	1.2207×10^{-5}	6.9337×10^{-6}	-1.3161×10^{-20}

Table 4: E_n at $x_n = 0.4$ against h

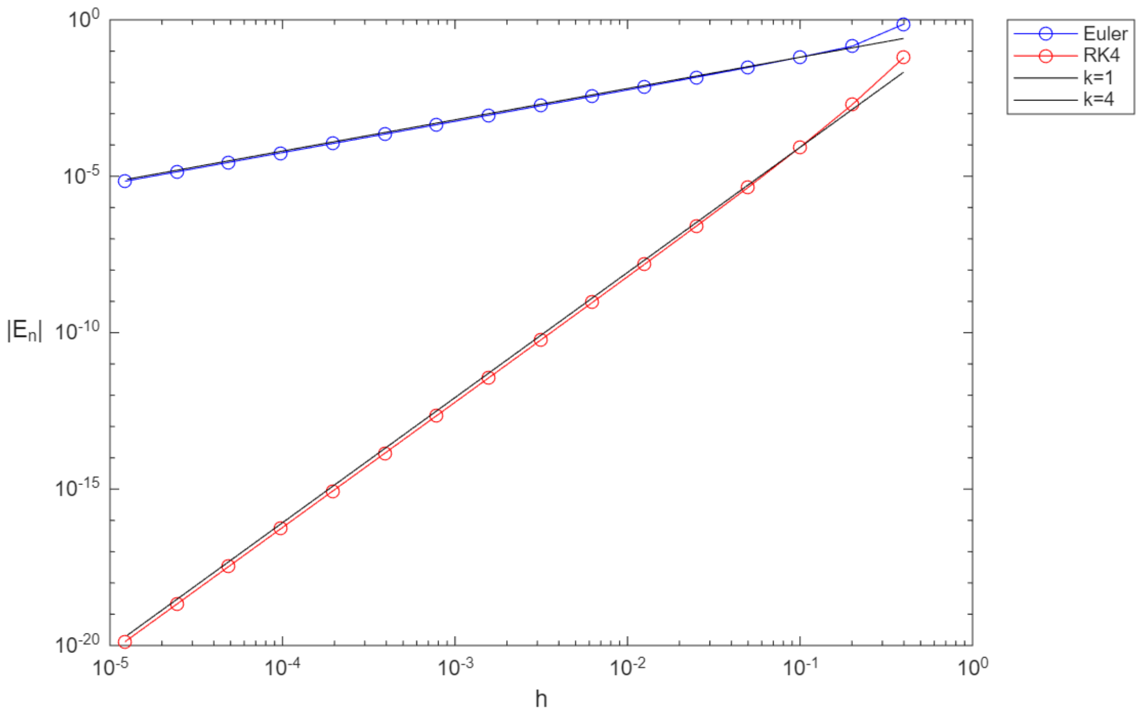

Figure 2: log-log graph of $|E_n|$ against h for the Euler (blue) and RK4 (red) methods

We know from theory that the global error E_n is $O(h)$ for the forward Euler method[1] and $O(h^4)$ for the RK4 method[2]. This is consistent with our results above; we see that the global error for the RK4 method decreases much more rapidly with h than the global error for the Euler method. Also, both curves are nearly straight lines in the log-log plot, indicating that the error can be closely approximated by power functions of the form Cx^k, where the exponent k is the gradient of the straight line in the log-log plot.

We can plot lines of gradient $k = 1$ and $k = 4$ (in black) on the log-log plot in Figure 2, and see that these closely approximate the error curves in both cases, agreeing with the theoretical accuracy of the two methods.

3 Numerical solutions of second-order ODEs

Question 5.

Suppose the system is lightly damped; that is, $0 < \gamma < 2$.

For the particular integral, we try a solution of the form $y_p(t) = A\sin(\omega t) + B\cos(\omega t)$. Substituting this into the differential equation (8) yields

$$-\omega^2[A\sin(\omega t) + B\cos(\omega t)] + \omega\gamma[A\cos(\omega t) - B\sin(\omega t)] + A\sin(\omega t) + B\cos(\omega t) = \sin(\omega t)$$

Equating the coefficients of $\sin(\omega t)$ and $\cos(\omega t)$ and solving the resulting system for A and B yields

$$A = \frac{1-\omega^2}{(1-\omega^2)^2 + \omega^2\gamma^2}$$
$$B = -\frac{\omega\gamma}{(1-\omega^2)^2 + \omega^2\gamma^2}$$

[1]Kharab, A. and Guenther, R.B., *An Introduction to Numerical Methods: A MATLAB Approach*, 4th Edition, page 352.
[2]Kharab, A. and Guenther, R.B., *An Introduction to Numerical Methods: A MATLAB Approach*, 4th Edition, page 361.

We see that we can rewrite $y_p(t) = A\sin(\omega t) + B\cos(\omega t)$ as $y_p(t) = R\sin(\omega t - \theta)$, where

$$R = \frac{1}{\sqrt{(1-\omega^2)^2 + \omega^2\gamma^2}}$$
$$\theta = \begin{cases} \arctan\left(\frac{\omega\gamma}{1-\omega^2}\right) & \omega \le 1 \\ \arctan\left(\frac{\omega\gamma}{1-\omega^2}\right) + \pi & \omega > 1 \end{cases}$$

For the complementary function, we observe that the homogenous equation has characteristic equation

$$\lambda^2 + \gamma\lambda + 1 = 0$$

which has roots $\lambda = \frac{-\gamma \pm i\sqrt{4-\gamma^2}}{2}$. The corresponding complementary function is then[3]

$$y_c(t) = \alpha e^{-\gamma t/2} \sin\left(\frac{\sqrt{4-\gamma^2}}{2} t\right) + \beta e^{-\gamma t/2} \cos\left(\frac{\sqrt{4-\gamma^2}}{2} t\right)$$

Now, observe that y_p has constant amplitude A, while $y_c \to 0$ as $t \to \infty$. Thus, we see that $y - y_p \to 0$, and so equation (9) holds with $A_s = R$ and $\phi_s = \theta$, i.e.

$$A_s = \frac{1}{\sqrt{(1-\omega^2)^2 + \omega^2\gamma^2}}$$
$$\phi_s = \begin{cases} \arctan\left(\frac{\omega\gamma}{1-\omega^2}\right) & \omega \le 1 \\ \arctan\left(\frac{\omega\gamma}{1-\omega^2}\right) + \pi & \omega > 1 \end{cases}$$

Question 6.

From our solution to Question 5, the analytic solution takes the form

$$\begin{aligned} y(t) &= y_c(t) + y_p(t) \\ &= e^{-\gamma t/2}\left[\alpha \sin\left(\frac{\sqrt{4-\gamma^2}}{2} t\right) + \beta \cos\left(\frac{\sqrt{4-\gamma^2}}{2} t\right)\right] + \frac{(1-\omega^2)\sin(\omega t) - \omega\gamma\cos(\omega t)}{(1-\omega^2)^2 + \omega^2\gamma^2} \end{aligned}$$

Imposing the initial conditions $y = \frac{\mathrm{d}y}{\mathrm{d}t} = 0$ at $t = 0$ given by Equation (14) yields

$$\beta - \frac{\omega\gamma}{(1-\omega^2)^2 + \omega^2\gamma^2} = 0$$
$$-\frac{\gamma\beta}{2} + \frac{\sqrt{4-\gamma^2}\alpha}{2} + \frac{\omega(1-\omega^2)}{(1-\omega^2)^2 + \omega^2\gamma^2} = 0$$

Solving for α and β yields

$$\alpha = \frac{\omega\gamma^2 - 2\omega(1-\omega^2)}{\sqrt{4-\gamma^2}[(1-\omega^2)^2 + \omega^2\gamma^2]}$$
$$\beta = \frac{\omega\gamma}{(1-\omega^2)^2 + \omega^2\gamma^2}$$

Then the analytic solution to Equation (8) with initial conditions given by Equation (14) is

$$\begin{aligned} y(t) =& \frac{1}{(1-\omega^2)^2 + \omega^2\gamma^2} e^{-\gamma t/2}\left[\frac{\omega\gamma^2 - 2\omega(1-\omega^2)}{\sqrt{4-\gamma^2}} \sin\left(\frac{\sqrt{4-\gamma^2}}{2} t\right) + \omega\gamma\cos\left(\frac{\sqrt{4-\gamma^2}}{2} t\right)\right] \\ &+ \frac{1}{(1-\omega^2)^2 + \omega^2\gamma^2}\left[(1-\omega^2)\sin(\omega t) - \omega\gamma\cos(\omega t)\right] \end{aligned} \tag{IV}$$

[3] Boyce, W.E. et.al., *Elementary Differential Equations and Boundary Value Problems*, 11th Edition, page 123

		$h = 0.4$		$h = 0.2$		$h = 0.1$	
t_n	$y(t_n)$	Y_n	E_n	Y_n	E_n	Y_n	E_n
0	0	0	0	0	0	0	0
0.1	0.00028104					0.00028111	7.143×10^{-8}
0.2	0.0021807			0.002183	2.2535×10^{-6}	0.0021808	1.1796×10^{-7}
0.4	0.016228	0.016297	6.9287×10^{-5}	0.016231	2.9436×10^{-6}	0.016228	1.4966×10^{-7}
0.8	0.10705	0.1071	4.4085×10^{-5}	0.10705	1.5788×10^{-6}	0.10705	7.9475×10^{-8}
1.2	0.27736	0.27735	-6.1869×10^{-6}	0.27736	6.7476×10^{-8}	0.27736	3.6949×10^{-8}
1.6	0.46319	0.46318	-1.7467×10^{-5}	0.46319	1.3904×10^{-6}	0.46319	1.7417×10^{-7}
2	0.56998	0.57002	3.616×10^{-5}	0.56999	5.9061×10^{-6}	0.56998	4.7753×10^{-7}
4	-0.4319	-0.43191	-1.3386×10^{-5}	-0.4319	-5.6396×10^{-6}	-0.4319	-4.9095×10^{-7}
6	0.34991	0.34989	-1.8776×10^{-5}	0.34991	3.6514×10^{-6}	0.34991	3.6105×10^{-7}
8	-0.3316	-0.33165	-5.1525×10^{-5}	-0.33161	-6.9362×10^{-6}	-0.3316	-5.2718×10^{-7}
10	0.27727	0.27742	1.4891×10^{-4}	0.27728	1.1698×10^{-5}	0.27727	7.7998×10^{-7}

Table 5: Y_n and E_n for $0 \leq t_n \leq 10$ with $h = 0.4$, 0.2 and 0.1

Comparing the local error $e_1^{(h)}$, we have

$$\frac{E_1^{(0.4)}}{E_1^{(0.2)}} = 30.7... \approx 2^5$$
$$\frac{E_1^{(0.2)}}{E_1^{(0.1)}} = 31.5... \approx 2^5$$

suggesting that $e_1^{(h)}$ is approximately proportional to h^5, which is consistent with the theory (which states that e_1 is $O(h^5)$ as $h \to 0$).

Comparing the global errors $E^{(h)}(t)$ at fixed t, say $t = 10$, we see that

$$\frac{E^{(0.4)}(10)}{E^{(0.2)}(10)} = 12.7...$$
$$\frac{E^{(0.2)}(10)}{E^{(0.1)}(10)} = 15.0... \approx 2^4$$

We see that the global error decays rapidly and nonlinearly with h but does not appear to be proportional to h^4 in the range of values investigated. This can be explained in the following two ways. We note that the theory (see Question (4)) states that the global error E_n is $O(h^4)$ as $h \to 0$, which bounds the error above by a function proportional to h^4, but does not mean the error itself must be proportional to h^4 as $h \to 0$. Also, this may be due to the range of values of h investigated; it is possible that the error may tend towards a function proportional to h^4 for smaller values of h.

Question 7.

We run our RK4 program for the given values of ω and γ, setting $h = 0.1$ in all cases. By comparing our numerical solutions with the corresponding analytic solutions, we see that at every time step $t_n \in [0, 40]$, this choice of h results in a global error of less than 0.5×10^{-5} for the given values of ω and γ, which is negligible on the scale of the graphs being plotted.

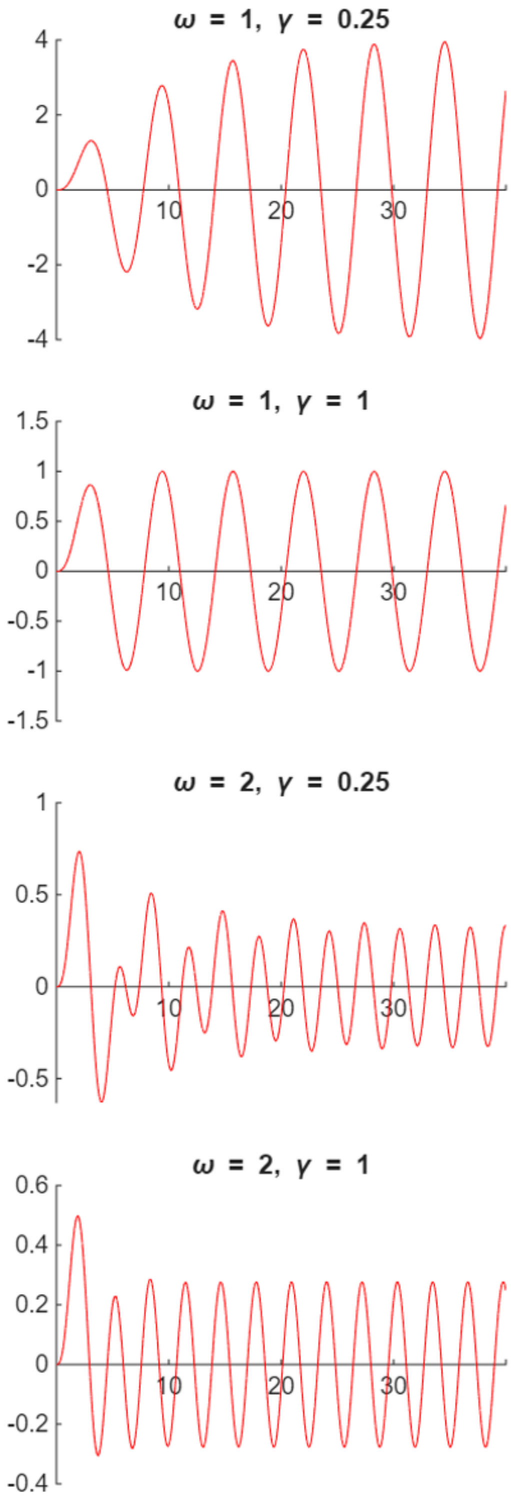

Figure 3: Y for $0 \leq t \leq 40$ for the given values of ω and γ, with $h = 0.1$

We make the following observations about the plots above.

1. Near $t=0$, Y oscillates about $Y=0$ with varying amplitude, but as $t\to\infty$, we see that Y approaches a solution which oscillates about $Y=0$ with (angular) frequency ω and some constant amplitude A_s, as predicted by Equation (9).

2. Within the given range of values of (ω,γ), the steady-state amplitude $A_s=A_s(\omega,\gamma)$ decreases with γ, from $A_s(1,0.25)\approx 4$ and $A_s(2,0.25)\approx 0.3$ to $A_s(1,1.9)\approx 0.5$ and $A_s(2,1.9)\approx 0.2$. From these data points, we also see that A_s appears to decrease with ω.

3. The rate at which the oscillations approach the steady-state amplitude increases with γ. In the case $(\omega,\gamma)=(1,0.25)$, the maxima nearest to $t=10$ has $|Y|\approx 0.7A_s$, whereas the corresponding maxima in the case $(\omega,\gamma)=(1,1.9)$ has $|Y|\approx A_s$. We see a similar trend by comparing $|Y|$ at the first maxima after $t=10$ in the cases $(\omega,\gamma)=(2,0.25)$ and $(\omega,\gamma)=(2,1.9)$. The relationship between this rate and ω is not clear from the two values of ω investigated.

4. Focusing on the cases $\gamma=0.25,0.5$ which approach the steady-state slowly, for $\omega=1$, the oscillation amplitude appears to increase monotonically with t, but for case $\omega=2$, the oscillation amplitude of y appears to oscillate about A_s for small t.

We can explain these observations by looking at the analytic solution to Equations (8) and (14). From the discussion in Question (6), the solution takes the form $y(t)=y_c(t)=y_p(t)$, where

$$\begin{aligned}
y_c(t) &= \frac{1}{(1-\omega^2)^2+\omega^2\gamma^2}e^{-\gamma t/2}\left[\frac{\omega\gamma^2-2\omega(1-\omega^2)}{\sqrt{4-\gamma^2}}\sin\left(\frac{\sqrt{4-\gamma^2}}{2}t\right)+\omega\gamma\cos\left(\frac{\sqrt{4-\gamma^2}}{2}t\right)\right] \\
y_p(t) &= \frac{1}{(1-\omega^2)^2+\omega^2\gamma^2}\left[(1-\omega^2)\sin(\omega t)-\omega\gamma\cos(\omega t)\right]
\end{aligned} \tag{V}$$

1. As in Question (5), we observe that $y_c(t)\to 0$ as $t\to\infty$, while $y_p(t)$ oscilllates about $y=0$ with constant amplitude $A_s=\frac{1}{\sqrt{(1-\omega^2)^2+\omega^2\gamma^2}}$ and frequency ω. This explains observation (1): for small t, $y_c(t)$ and $y_p(t)$ are comparable in magnitude, so the behaviour of y near $t=0$ (the transient response) is determined by both y_c (which oscillates with decaying amplitude) and y_p, but as t increases, $y_c(t)$ decays exponentially and becomes negligible compared to $y_p(t)$, so the behaviour of y as $t\to\infty$ (the steady-state) is determined entirely by y_p. Physically, we see that the oscillatory mode with frequency $\frac{\sqrt{4-\gamma^2}}{2}$ loses energy due to the damping term $\gamma\frac{\mathrm{d}y}{\mathrm{d}t}$ in Equation (8), causing its amplitude to decay, while the oscillatory mode with frequency ω is driven by the forcing term $\sin(\omega t)$ and thus remains at a constant amplitude.

2. From the expression for A_s above, we can see that A_s decreases with γ, explaining observation (2). Physically, we see that increasing γ increases resistive forces in the system, which oppose motion and decrease the steady-state amplitude. Also, in the range $\omega\geq 1$ under investigation, A_s decreases with ω, as observed earlier. Heuristically, the damping term $\gamma\frac{\mathrm{d}y}{\mathrm{d}t}$ is proportional to the velocity $\frac{\mathrm{d}y}{\mathrm{d}t}$, and increasing ω causes the magnitude $\frac{\mathrm{d}y}{\mathrm{d}t}$ to increase relative to y, resulting in increased resistive forces and thus reduced amplitude.

3. Also, from the expression (V) for $y_c(t)$, we see that the complementary function has amplitude $A(t)$ that decays as $A(t)=Ce^{-\gamma/2}$, where $C=C(\omega,\gamma)$ is some constant independent of t. Then, the time taken for the amplitude of y_c to decay by a factor of $1/e$ (the relaxation time) is given by $\tau=\frac{2}{\gamma}$, which decreases with γ and is independent of ω, explaining observation (3). Physically, the greater the magnitude of the damping force, the faster energy is dissipated, causing the amplitude of that oscillatory mode to decay faster.

4. Finally, we note that for small values of γ, y_c oscillates with frequency $\frac{\sqrt{4-\gamma^2}}{2}=1-\frac{\gamma^2}{8}+...\approx 1$. Then, when the driving force has $\omega=1$, $y_p(t)$ and $y_c(t)$ oscillate with approximately the same frequency.

Then we see that $\text{sgn}(y_c(t))$ will be the same at every maxima of $y_p(t)$, and so the amplitude of the oscillations in y will change monotonically as $y_c(t) \to 0$. When the driving force has $\omega = 2$, the frequency of $y_p(t)$ is approximately twice that of $y_c(t)$, so $\text{sgn}(y_c(t))$ alternates between ± 1 at consecutive maxima of $y_p(t)$, causing the oscillation amplitude of y at small t to oscillate about A_s. This explains observation (4).

Question 8.

We can rewrite Equation (15) as

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + \delta^3 y^2 \frac{\mathrm{d}y}{\mathrm{d}t} + y = \sin(t) \tag{VI}$$

For each value of δ, we start with $h_0 = 0.08$ and use our program to calculate $Y^0(t_n)$ at the points $t_n = 2, 4, ..., 60$. We then take $h_{k+1} = \frac{1}{2}h_k$ and repeat. We stop once we reach a value h_k such that $Y^{k-2}(t_n)$, $Y^{k1}(t_n)$ and $Y^k(t_n)$ are identical (to three significant figures) at the points $t_n = 2, 4, ..., 60$. Since the RK4 method is fourth-order, this process should give us a satisfactory approximation to the exact solution y.

Running our program, we see that this occurs for $h = 0.01$ for all of the given values of δ, so we proceed with this value of h.

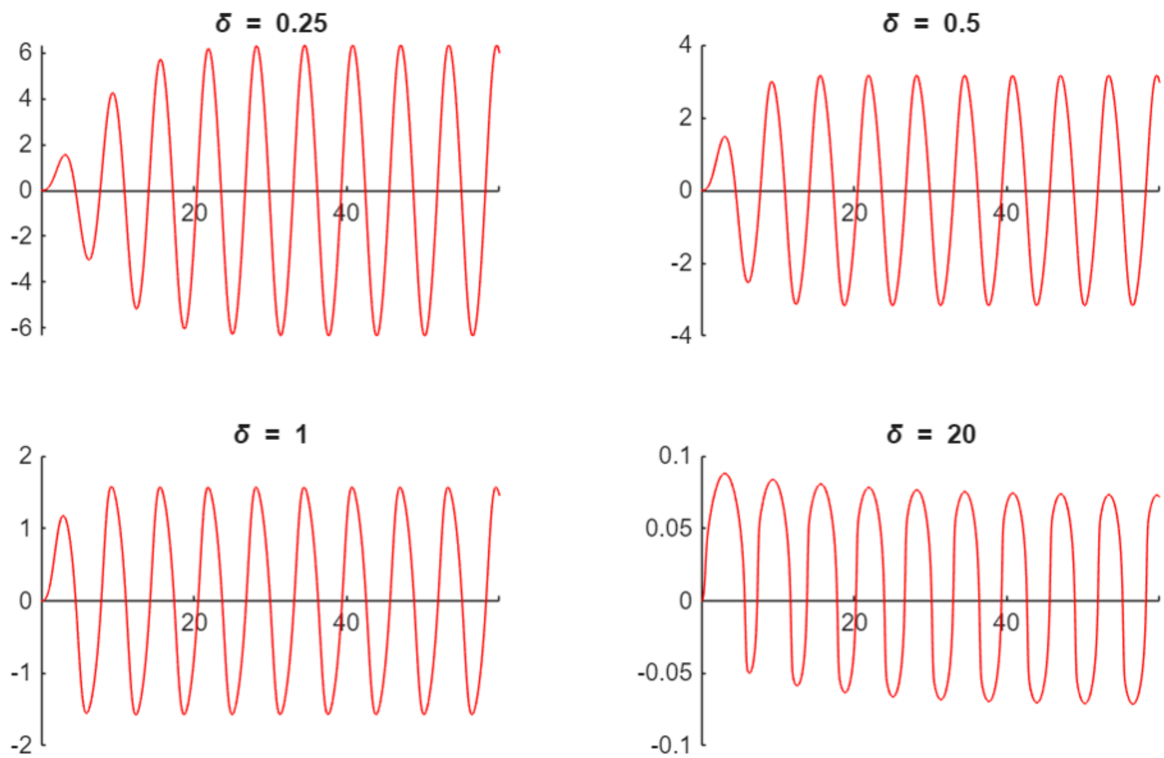

Figure 4: Y for $0 \leq t \leq 60$ for the given values of δ, with $h = 0.01$

We make the following observations about the plots above.

1. As before, near $t = 0$, Y oscillates about $Y = 0$ with varying amplitude, but as $t \to \infty$, Y approaches a solution which oscillates about $Y = 0$ with (angular) frequency $\omega = 1$ and some constant amplitude A_s. Physically, the explanation is the same as in Question (7); the oscillatory mode with frequency $\omega = 1$ is driven by the forcing term, while all other modes decay due to energy loss from the damping term.

2. Within the range of values of values of δ investigated, the steady-state amplitude A_s decreases as δ increases. Also, we see that the steady-state amplitude A_s for the cases $\delta = 0.25, 0.5, 1.0$ is greater than the corresponding steady-state amplitudes in the linear damping case with $\gamma = 0.25, 0.5, 1.0$.
3. The rate at which the oscillation amplitude approaches the steady-state amplitude A_s increases with δ. We see that the oscillation amplitude after one period has reached roughly 70% of the steady-state amplitude in the case $\delta = 0.25$, but is nearly indistinguishable from the steady-state amplitude in the case $\delta = 1.0$. Physically, the explanation is the same as in the case of linear damping; the greater the magnitude of the damping force, the faster the complementary function decays.
4. The steady-state solution is no longer exactly sinusoidal, which is most evident in the plot with $\delta = 20$; the gradient of Y appears to be much steeper near $Y = 0$ and less steep near the extrema of Y, as compared to a sinusoidal function.

For sufficiently small values of δ, Equation (15) has a 2π-periodic solution of the form

$$y^{(p)} = \sum_{n=-1}^{\infty} \delta^n y_n^{(p)}(t)$$

where each $y_n(t)$ is periodic in t with period 2π, and

$$y_{-1}^{(p)}(t) = A\cos(t)$$
$$y_0^{(p)}(t) = B\sin(t) + C\sin(3t)$$

Substituting the expression for $y^{(p)}$ into Equation (15) and comparing coefficients of δ^0, we see that

$$A = -\sqrt[3]{4}$$

Then since δ is small, $y^{(p)} = \delta^{-1}y_{-1}^{(p)} + O(1) \approx \delta^{-1}y_{-1}^{(p)}$. Also, since this solution is 2π-periodic (and so has the same period as the forcing term $\sin(t)$), this solution determines the steady-state behaviour of the system. We can superpose the function $\hat{y} = \delta^{-1}y_{-1}^{(p)}$ on the numerical output of our program for $\delta = 0.25, 0.5$ and see that the numerical solution Y does indeed appear to converge to $\hat{y}$ for large t.

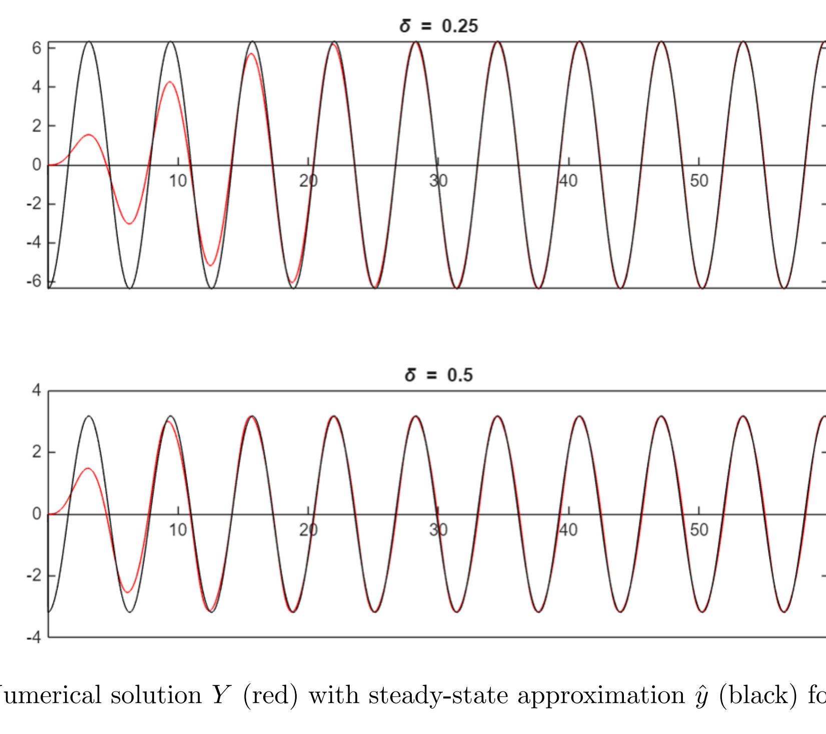

Figure 5: Numerical solution Y (red) with steady-state approximation $\hat{y}$ (black) for $\delta = 0.25, 0.5$

For large values of δ, we can instead write

$$y^{(p)} = \sum_{n=-\infty}^{-1} \delta^n y_n^{(p)}$$

Then $y^{(p)} = \delta^{-1} y_{-1}^{(p)} + O(\delta^{-2}) \approx \delta^{-1} y_{-1}^{(p)}$. Substituting this expression into Equation (15) and comparing coefficients of δ^0, we see that we get

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{3} y_{-1}^3\right) &= \sin(t) \\ \implies y_{-1}(t) &= -\sqrt[3]{3\cos(t) + C} \end{aligned}$$

where C is the constant of integration. Physically, we expect the system to oscillate about $y = 0$ at steady-state, so we take $C = 0$.

Superposing the function $\hat{y}(t) = -\delta^{-1}\sqrt[3]{3\cos(t)}$ on the numerical output of our program for $\delta = 20$, we see that the numerical solution Y does indeed appear to converge to $\hat{y}$ for large t, which explains the non-sinusoidal steady-state behaviour of the system for large δ (observation (4)).

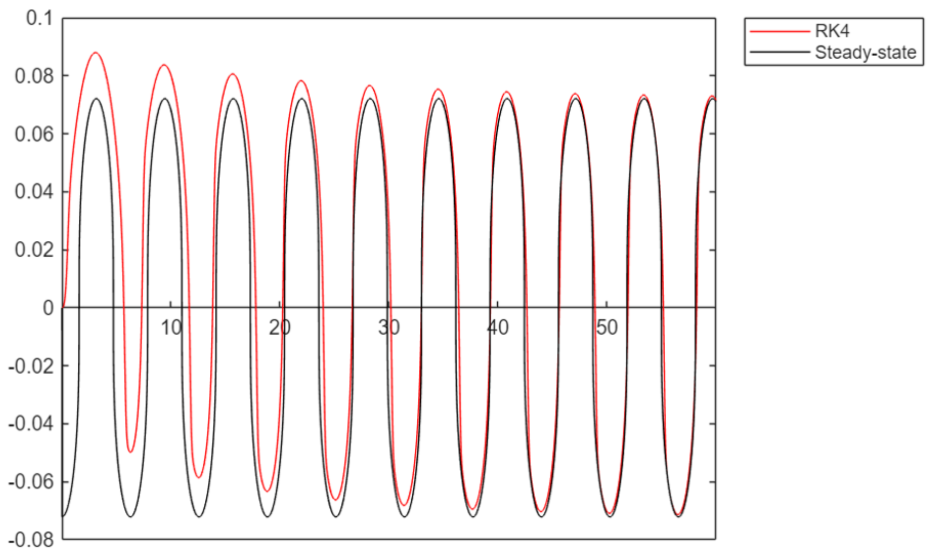

Figure 6: Numerical solution Y (red) with steady-state approximation $\hat{y}$ (black) for $\delta = 20$

4 Program Listings

Euler Method

```
%Set initial values x_0, Y_0, step size h, and initialise n
xn = 0;
Yn = 0;
h = 0.6;
n = 0;

%Set initial values for analytic value yn, global error En, Gn, Gn-1
yn = 0;
En = 0;
Gn = 0; %G_(n-1)
Gnm = 0; %G_n

disp(['$' num2str(n) '$ & $' num2str(xn) '$ & $' num2str(Yn) ...
    '$ & $' num2str(yn) '$ & $' num2str(En) '$ & $' ...
    num2str(Gn) '$ & $' num2str(Gn-Gnm) '$ \\'])

while n < 15
    %Increment loop counter
    n = n+1;

    %Update Yn
    Yn = Yn + h.*f(xn,Yn);

    %Update xn
    xn = xn + h;

    %Update yn (analytic solution)
    yn = g(xn);

    %Calculate En
    En = Yn - yn;

    %Set Gnm
    Gnm = Gn;

    %Calculate Gn
    Gn = log(abs(En)) ./ h;

    disp(['$' num2str(n) '$ & $' num2str(xn) '$ & $' num2str(Yn) ...
    '$ & $' num2str(yn) '$ & $' num2str(En) '$ & $' ...
    num2str(Gn) '$ & $' num2str(Gn-Gnm) '$ \\'])
end
```

RK4 Method

```
%Set initial values x_0, Y_0, step size h, and initialise n
xn = 0;
Yn = 0;
h = 0.2;
n = 0;

%Set initial values for analytic value yn, global error En
yn = 0;
En = 0;

%Initialise variables k1,k2,k3,k4 used in RK4
k1 = 0;
k2 = 0;
k3 = 0;
k4 = 0;

disp([num2str(Yn)])

while xn < 4
    n = n+1; %Increment loop counter

    k1 = h.*f(xn,Yn);     %Update k1
    k2 = h.*f(xn + h./2, Yn + k1 ./2);     %Update k2
    k3 = h.*f(xn + h./2, Yn + k2 ./2);     %Update k3
    k4 = h.*f(xn+h, Yn + k3);     %Update k4

    Yn = Yn + (k1 + 2.*k2 + 2.*k3 + k4)./6;     %Update Yn
    xn = xn + h;     %Update xn
    yn = g(xn);     %Update yn (analytic solution)
    En = Yn - yn;     %Calculate En

    disp([num2str(Yn)])
end
```

RK4 Method for first-order systems of equations

```
%Set initial values t_0, Y_0, step size h, and initialise n
tn = vpa(0);
Yn = vpa(zeros(2,1));
h = vpa(0.4);
n = 0;

%Set initial values for analytic value yn, global error En
yn = vpa(0);
En = 0;

%Initialise dummy variables k1,k2,k3,k4 used in RK4
k1 = zeros(2,1);
k2 = zeros(2,1);
k3 = zeros(2,1);
k4 = zeros(2,1);

disp(['$' num2str(double(tn)) '$ & $' num2str(double(Yn(1))) ...
    '$ & $' num2str(double(En)) '$ \\'])

while n < 25
    n = n+1;     %Increment loop counter

    k1 = h.*f1(tn,Yn);     %Update k1
    k2 = h.*f1(tn + h./2, Yn + k1 ./2);     %Update k2
    k3 = h.*f1(tn + h./2, Yn + k2 ./2);     %Update k3
    k4 = h.*f1(tn+h, Yn + k3);     %Update k4

    Yn = Yn + (k1 + 2.*k2 + 2.*k3 + k4)./6;     %Update Yn
    tn = tn + h;     %Update xn
    yn = g1(tn);     %Update yn (analytic solution)
    En = Yn(1) - yn;     %Calculate En

    disp(['$' num2str(double(tn)) '$ & $' num2str(double(yn)) ...
        '$ & $' num2str(double(Yn(1))) ...
    '$ & $' num2str(double(En)) '$ \\'])
end
```